A computer-assisted three-state counterexample to a nonlinear phase-lag bound

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Abstract

A phase-lag bound was proved for periodically driven two-state Markov systems and conjectured for arbitrary finite networks by replacing the single rate prefactor with the smallest edge prefactor. We disprove the conjecture already on the complete three-state graph. The counterexample retains local detailed balance, all three edges, time-independent positive prefactors, and a two-harmonic energy drive. A computer-assisted residual argument encloses the exact periodic state uniformly in time. Outward-rounded ball arithmetic certifies that both phase integrals are positive while the proposed bound margin is strictly negative. Thus the phase exceeds the proposed upper bound. The result shows that the slowest bare edge prefactor alone does not control nonlinear multistate phase lag; no replacement bound is claimed.

Keywords: Stochastic thermodynamics; Dynamical processes; Exact results; Stationary states

1 Introduction

Periodic driving produces an intrinsic lag between a Markov system and its changing energy landscape. Degünther, Koyuk and Seifert introduced a scalar phase for this lag and proved a nonlinear upper bound for two-state systems [1]. For a general network with heterogeneous edge prefactors, they proposed the extension

$$\Delta\varphi \leq \arctan\left(\frac{\Omega}{k_{0,\min}}\right), \tag{1}$$

supported by random three-state calculations. The same statement was still described as a conjecture in the later thesis of Degünther [2]. The two-state proof does not directly extend because a multistate network has several energy differences, rate splittings, and relaxation channels.

Here we give one fully specified complete-three-state system for which Eq. (1) fails. A floating-point solve is used only to select a comparison path. The mathematical conclusion follows from interval enclosures of its continuum residual, an analytic contraction estimate for the exact period map, and interval evaluation of the observables. This separates discovery from the certificate and rules out an apparent violation caused by time discretization or an imperfectly periodic trajectory.

2 Model and the proposed bound

Consider states $i \in \{1, 2, 3\}$, with all three undirected edges present and $k_{\mathrm{B}}T = 1$. For $i < j$, let

$$k_{ij}(t) = k_{0,ij} \exp[\alpha_{ij}(E_i(t) - E_j(t))], \tag{2}$$

$$k_{ji}(t) = k_{0,ij} \exp[-(1 - \alpha_{ij})(E_i(t) - E_j(t))], \tag{3}$$

where $k_{0,ij} > 0$ and $0 \le \alpha_{ij} \le 1$ are constant. Hence $k_{ij}/k_{ji} = \exp(E_i - E_j)$. The energies have a finite Fourier expansion,

$$E_i(t) = E_i^0 + \sum_{n=1}^{M} \left[c_{ni} \sin(n\Omega t) + d_{ni} \cos(n\Omega t) \right]. \tag{4}$$

The probability column p obeys

$$\dot{p}_i = \sum_{j \neq i} (p_j k_{ji} - p_i k_{ij}), \qquad p(t+T) = p(t), \quad T = \frac{2\pi}{\Omega}. \tag{5}$$

Strictly positive continuous rates make the one-period stochastic matrix positive; therefore its normalized fixed point exists, is unique, and is componentwise positive.

Define the period averages

$$A = \frac{1}{T} \int_0^T p(t) \cdot \ddot{E}(t)\, \mathrm{d}t, \qquad B = \frac{1}{T} \int_0^T p(t) \cdot \dot{E}(t)\, \mathrm{d}t. \tag{6}$$

On the nondegenerate branch $A > 0, B > 0$, the multistate phase of Ref. [1] is

$$\Delta\varphi = \arctan\left(\frac{\Omega B}{A} \right). \tag{7}$$

Because arctan is increasing and $k_{0,\min} > 0$, Eq. (1) is then equivalent to

$$A - k_{0,\min} B \ge 0. \tag{8}$$

3 Exact counterexample

Set $M = 2$, $\Omega = 6$, and use the exact terminating decimals

edge ij	12	13	23
$k_{0,ij}$	1	300	8000
α_{ij}	0.01	0.83	0.93

(9)

with $\theta = \Omega t$ and

$$E_1(\theta) = -13 + 0.8 \sin\theta + 1.1 \sin 2\theta - 1.7 \cos\theta + 6.8 \cos 2\theta, \tag{10}$$

$$E_2(\theta) = -23 + 0.85 \sin\theta + 6.1 \sin 2\theta + 1.7 \cos\theta - 3 \cos 2\theta, \tag{11}$$

$$E_3(\theta) = 0. \tag{12}$$

Every edge is retained, every prefactor is fixed and positive, and every splitting parameter lies strictly inside its allowed interval. In particular, $k_{0,\min} = 1$.

Our central result is the following.

Proposition. *The exact periodic state of Eqs.* (2)–(12) *satisfies* $A > 0$, $B > 0$, *and* $A - B < 0$. *Consequently it violates Eq.* (1).

The proof is computer-assisted; all discretization and rounding errors used in the conclusion are enclosed below.

4 Residual certificate

4.1 Periodic comparison path

In the phase variable, Eq. (5) is

$$p'(\theta) = G(\theta)p(\theta), \qquad G(\theta) = L(\theta)/\Omega, \tag{13}$$

where the columns of L sum to zero. A numerical Radau solve supplies values and slopes on 2048 uniform segments. The final value and slope are set equal to the initial ones, and cubic Hermite interpolation defines $\widehat{p}_1, \widehat{p}_2$; we set $\widehat{p}_3 = 1 - \widehat{p}_1 - \widehat{p}_2$. Thus the comparison path is exactly normalized, periodic, and C^1, regardless of the accuracy of the proposal solve. Each binary floating-point node and slope is converted to its exact integer ratio before evaluation in Arb ball arithmetic [3].

Let

$$r(\theta) = \widehat{p}'(\theta) - G(\theta)\widehat{p}(\theta). \tag{14}$$

Each Hermite segment is divided into 1024 closed subintervals. At 256-bit working precision, outward-rounded interval evaluation on all 2,097,152 boxes gives

$$R := \int_0^{2\pi} \|r(\theta)\|_1 \, \mathrm{d}\theta \le 8.40694407275998 \times 10^{-5}. \tag{15}$$

4.2 From residual to the exact periodic state

For a continuous-time Markov generator with directed rates q_{ij}, the standard coupling estimate contracts zero-sum signed measures in ℓ^1 at the instantaneous rate

$$\gamma = \min_{i \neq j} \left(q_{ij} + q_{ji} + \sum_{\ell \neq i,j} \min(q_{i\ell}, q_{j\ell}) \right). \tag{16}$$

Indeed, for a pair of starting states i, j, the displayed rate is the sum of direct jumps between the pair and the maximum common jump rate into every third state; an infinitesimal coupling followed by integration gives the contraction. For the three unordered pairs, the bracket is bounded below respectively by k_{12}, k_{31}, and k_{32} in physical time. Direct triangle bounds on Eqs. (10) and (11) give

$$E_1 - E_2 \ge -8.25, \qquad E_1 \le -2.6, \qquad E_2 \le -11.35. \tag{17}$$

It follows from Eqs. (2) and (3) that $k_{12} \ge \exp(-33/400)$, whereas $k_{31} > 1$ and $k_{32} > 1$. The exact one-period map therefore contracts the zero-sum subspace by at most

$$q = \exp\left[-\frac{2\pi}{6} \mathrm{e}^{-33/400} \right] \le 0.381260853664531 < 1. \tag{18}$$

Put $e = p - \widehat{p}$. Variation of constants, periodicity of both paths, and Eq. (18) imply

$$\|e(0)\|_1 \le \frac{R}{1-q}, \qquad \sup_{0 \le \theta \le 2\pi} \|e(\theta)\|_1 \le \frac{R}{1-q} + R \le 2.19941627295248 \times 10^{-4}. \tag{19}$$

This step connects the finite comparison path to the unique exact periodic state; no numerical monodromy matrix or numerical ODE tolerance is assumed to be rigorous.

4.3 Observable enclosure

The Hermite cubics are integrated in closed form against the two Fourier harmonics, with all evaluations enclosed by Arb. For the comparison path this gives

$$\widehat{A} = 0.4997274966496502\ldots, \qquad \widehat{B} = 6.1308368530420400\ldots. \tag{20}$$

In θ units the exact coefficients give

$$\max_i \|\partial_\theta E_i\|_\infty \le 20.75, \qquad \max_i \|\partial_\theta^2 E_i\|_\infty \le 38.95. \tag{21}$$

Combining these bounds with Eq. (19) yields

$$|A - \widehat{A}| \le \Omega^2(38.95) \sup_\theta \|e(\theta)\|_1 \le 0.308402149793396, \tag{22}$$

$$|B - \widehat{B}| \le \Omega(20.75) \sup_\theta \|e(\theta)\|_1 \le 0.027382732598259. \tag{23}$$

The final outward-rounded enclosures are

$$A \in [0.1913253462459419,\ 0.8081296470533585], \tag{24}$$

$$B \in [6.1034541204110998,\ 6.1582195856729802], \tag{25}$$

$$A - B \in [-5.966894239630766,\ -5.295324473154014]. \tag{26}$$

The first two lower endpoints establish the stipulated positive branch; the negative upper endpoint in the third line proves the proposition. A separate run with 4096 Hermite segments, 512 residual subboxes per segment, and 320-bit precision again certifies all three strict signs.

5 Physical meaning and limitations

The counterexample is not produced by deleting a slow edge or by changing the phase convention. It is a complete, irreducible local-detailed-balance network on the same positive branch for which the bound was proposed. The witness combines heterogeneous prefactors, asymmetric rate splittings, and two drive harmonics; the certificate does not identify any one of these features as necessary. What it establishes is narrow but decisive: the slowest bare prefactor is insufficient as a universal nonlinear multistate phase control.

We do not classify the set of violating drives, propose an alternative network timescale, or address a different phase observable. The example also does not invalidate the proven two-state results of Ref. [1]. Finding a replacement bound may require information from the full generator, its time-dependent contraction, or network topology rather than one edge prefactor.

Data and code availability

The exact-decimal parameters, the certificate program, the primary 256-bit output, a changed-partition 320-bit replay, and environment instructions accompany this draft. The certificate program regenerates the comparison path and performs every interval evaluation used in Eqs. (15)–(26).

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